

Project Catalyst



A Go-to-Live Strategy for Launching
AI-Powered Personalized Offers

"To enhance the American Express digital presence, evolving it from a platform of premier service into an intelligent engine for opportunity that anticipates and rewards every member."

***Goal:** Drive a **12-to-18% lift** in offer activation and an **8-to-12% increase** in incremental card spend post-launch.*



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Product Architecture

DATA SOURCES

- **Internal:** Transaction History (SQL DBs), CRM Data (Salesforce), Real-time Clickstream (Adobe/Google Analytics), Card Member Profile
- **External:** Open Banking APIs (e.g., Plaid) for a holistic financial view, Partner Data Feeds (e.g., airline sales, hotel promotions).

PERSONALIZATION ENGINE

- **Feature Store (Tecton/Feast):**
 - Real-time ML features
- **ML Models (SageMaker/Vertex AI):**
 - Collaborative Filtering: "Users like you also liked..."
 - Propensity Models (XGBoost)
 - LLMs (Google/OpenAI)
- **Rules Engine:**
 - Business logic & eligibility
 - Managed via a Low-Code portal for Marketing & Product teams.

DATA INGESTION

- **Real-time Stream:** Apache Kafka or AWS Kinesis to capture live user interactions.
- **Batch Ingestion:** Using Airflow for periodic data syncs.
- **Central Data Lakehouse:** Snowflake or Databricks. This provides a single source of truth, combining structured and unstructured data for both BI and ML.

DELIVERY LAYER

- **Offer API (REST):** A low-latency microservice that serves the ranked list of offers to the frontend.
- **Frontend:** The AMEX website, which calls the API upon user login or page refresh.

Backend Logic

Our AI Logic: The Personalized Value Proposition (PVP) Framework

WHO IS THE USER?

Card Type: Platinum, Gold, Blue Cash.

Spend Profile: High-value traveler, daily commuter, online shopper.

Historical Engagement: Which offers have they activated or ignored in the past?

WHAT ARE THEY DOING?

Clickstream: Viewing flights to London? Browse dining offers?

Real-time Context: Time of day, device type, location (with consent).

Search Queries: What are they explicitly looking for on the site?

RELEVANT EXTERNAL FACTORS

Partner Data: A partner hotel in London has a flash sale.

Market Trends: Increased interest in 'work from home' tech.

Life Events (Inferred): A large purchase at a furniture store might suggest a recent move.

DISCOVER

Goal: Broaden member horizons with new experiences they'll love.

How It Works: *Collaborative Filtering* models analyze a user's "taste profile" against similar members to suggest new merchants and offers.

Connects to Data: Primarily uses "Historical Engagement" and "Spend Profile."

OPTIMIZE

Goal: Make members' existing spending more rewarding and efficient.

How It Works: *Propensity Models (XGBoost)* predict the likelihood of a user activating an offer on things they already buy, prioritizing savings and rewards.

Connects to Data: Primarily uses "Clickstream" and "Transaction History."

ELEVATE

Goal: Partner in our members' aspirational life moments.

How It Works: *LLMs and Inference Models* detect triggers like travel planning to deliver exclusive benefits with personalized, empathetic messaging.

Connects to Data: Primarily uses "Search Queries," "Life Events (Inferred)," and "Partner Data."

LOW-CODE CAMPAIGN PORTAL

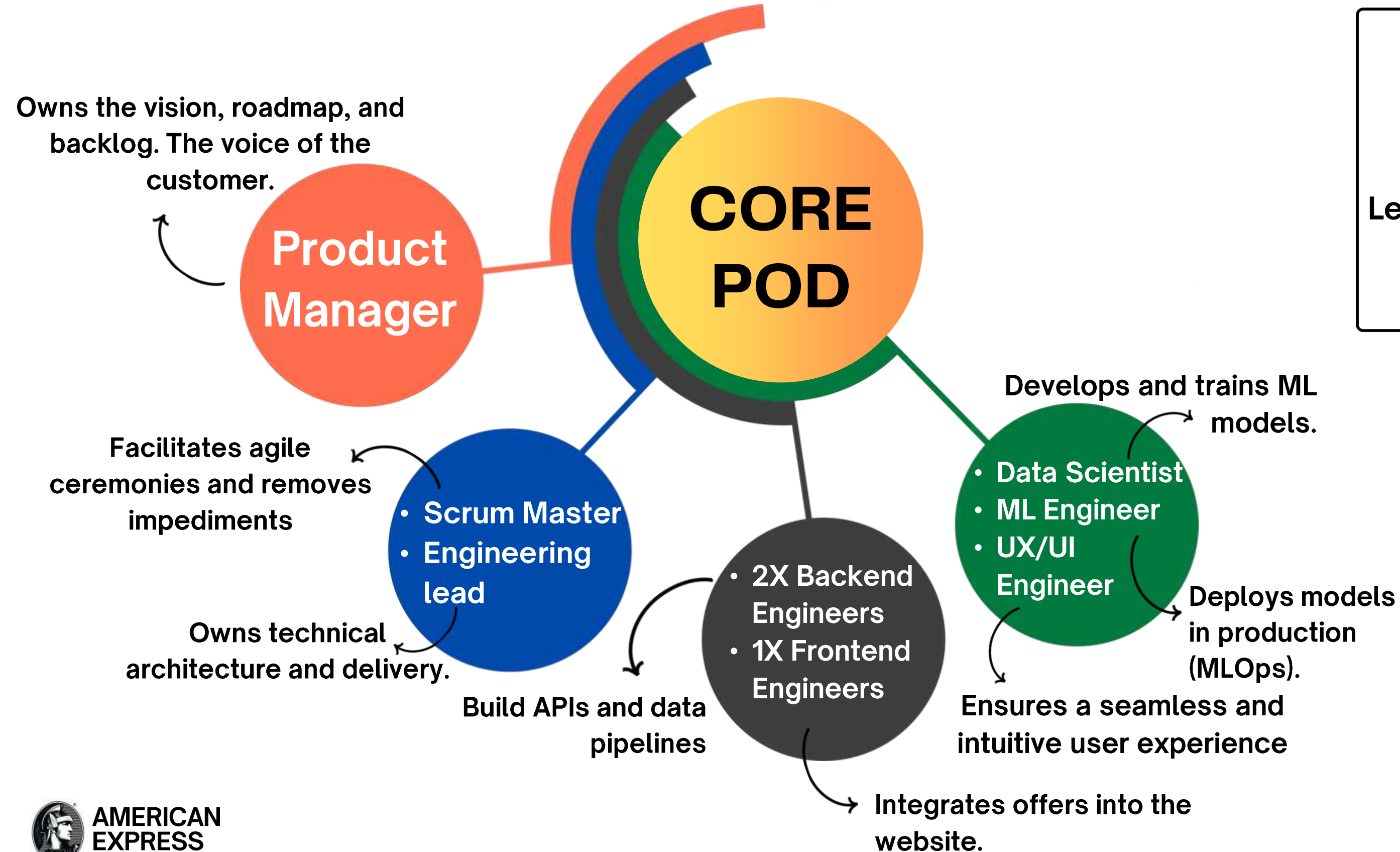
Our intuitive *low-code portal* empowers Marketing and Product teams to accelerate speed-to-market by directly managing campaigns, running A/B tests, and applying strategic boosts to the AI's logic. This frees up engineering resources to focus on core model innovation.



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Team Structure

The "Catalyst" Pod:
A Cross-Functional, Mission-Oriented Team



Support Functions

Marketing: For offer strategy and campaign alignment.
Legal & Compliance: To ensure data privacy and regulatory adherence (Privacy by Design).

Collaboration Model

Sprints: 2-week cycles.
Tools: **Jira** for backlog management, **Confluence** for documentation, **Figma** for design, **GitHub** for code.

Launch Plan

Deployment Strategy:

We will use **Canary Releases** and feature flags. This allows us to activate the personalization engine for a small subset of users first, monitor for errors, and roll back instantly if needed, ensuring zero downtime.

Development & Alpha Weeks 1-6

Users: Internal AMEX Employees (Opt-in)

Phase 0

Validate technical stack & data pipelines.

Closed Beta Weeks 7-10

Users: 1% of engaged users (by invitation)

Phase 1

Gather qualitative feedback & A/B test models.

Gradual Rollout (GA) Weeks 11-14

Ramp up: 10% → 30% → 75% → 100% of users

Phase 2

Monitor system performance and business KPIs at scale.

Optimization Week 15 onwards

All Users

Phase 3

Continuously retrain models and add new features.



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Success Metrics

ENGAGEMENT

- Offer Click-Through Rate (CTR)
- Offer Activation Rate (North Star Metric)

BUSINESS IMPACT

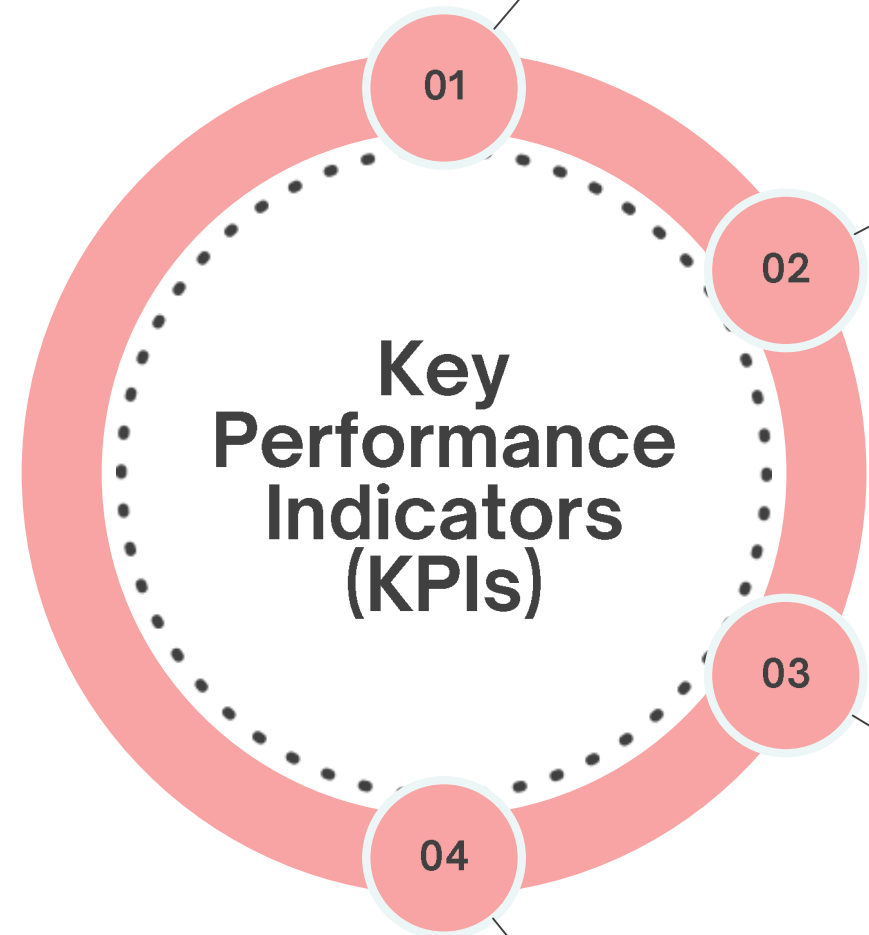
Incremental Lift in Card Spend on activated offers.

CUSTOMER SATISFACTION

Net Promoter Score (NPS) surveys. Reduction in offer-related customer service calls.

OPERATIONAL HEALTH

API latency (<200ms). Model prediction accuracy and drift monitoring.



Risk Mitigation

DATA PRIVACY BREACH

Mitigation: Adherence to GDPR/CCPA, data encryption at rest and in transit, Privacy by Design consultations with Legal.

PERSONALIZATION ERROR

Mitigation: Human-in-the-loop review for sensitive offers, robust user feedback mechanism

MODEL BIAS

Mitigation: Regular audits for bias using toolkits like AIF360. Ensure training data represents our diverse user base

SCALABILITY FAILURE

Mitigation: Cloud-native, auto-scaling architecture (Kubernetes), rigorous load testing before each phase, and proactive monitoring with Datadog/New Relic.



Future Roadmap

The Next Frontier: Proactive & Conversational Value

Our launch is just the beginning.

The “Catalyst” architecture enables a future where:

- **Hyper-Contextual Offers:** Real-time, location-aware deals —> exactly when and where users need them.
- **Omnichannel Personalization:** Seamless outreach via app, email, and SMS —> tailored to each user segment.
- **Conversational AI:** Smart assistant to help users discover and manage offers effortlessly.
- **Predictive Budgeting:** Forecast-based suggestions that align with spend patterns.
- **Partner Ecosystem Expansion:** Evolving our *internal low-code portal* into a no-code, self-serve platform for trusted merchant partners to directly configure and launch their Amex Offers, creating a scalable, automated pipeline of value for our members.